We Are This Pattern:
What Stays Alive When Everything Changes

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Abstract

This paper explains one idea: we are not the stuff we are made of, we are the way that stuff holds together as one loop. The loop borrows energy that must spread, uses it to stay itself while changing, and can be checked by a record kept outside the storyteller. When the loop holds, it is alive in the sense used here: it keeps borrowing the same energy to stay itself while changing—whether it looks like a bacterium, a human, a price, a law, or a star. Law and star cases are structural analogies with matched bookkeeping, not causal proof (see Limitations). When the loop scatters, new loops begin borrowing the same energy to look like new things. The paper shows five jobs the loop must do at once, what happens when any one is missing, and one test that would prove this idea false. Key estimates below are reported with working thresholds and should be read as consistent-with rather than proof (see Methods and Limitations).

1 We Are Not Stuff, We Are How Stuff Holds Together

Every day your body replaces most of its atoms. The water in you turns over in about ten days. Your blood in about one hundred and twenty days. Even the connections in your brain that feel most like you change a few percent in a single day. If you were the atoms, you would be someone else every week.

You are not the atoms. You are the way the atoms hold together. Think of a whirlpool, not a rock. A rock stays because it is hard. A whirlpool stays because water keeps moving through it in the same pattern. Stop the water and the whirlpool vanishes, even though the water is still there. Your pattern is like that whirlpool: water through, shape stays because motion makes more motion.

A famous thought experiment makes this clear. Imagine lightning by chance creates an exact copy of you in a swamp, atom for atom, with all the same stuff but none of the history of how that pattern was maintained. That copy would have everything, yet on the next step it would drift and forget about thirty percent, while you would hold about thirty percent and drift about thirty percent the other way. Same stuff, different history, different behavior. Stuff is not enough. How the stuff was held matters. This is why philosophers who study identity stress that memory and story, not atoms, carry who we are (Locke 1689; Parfit 1984; Schechtman 1996; Olson 1997; Zahavi 2005; Gallagher 2000) [18–23].

We see the same in our own work. We gave two artificial minds the exact same memories and asked each who the real one was. They answered in ways only six percent alike. Identical data produced genuinely different identity claims. The data did not decide. How the data was held decided. This matches what memory-system studies find: persona drifts more than thirty percent in eight to twelve turns, and larger models drift worse (Choi et al. 2024) [14]; role alone is unstable (Shanahan et al. 2023) [13].

2 A Day In One Loop

At any moment you are not just a brain. You are a whole state: body, tiredness, feeling, mood, what you need, what you are paying attention to, what you remember, what you expect, habits, relationships, your picture of yourself, your goals, your values. The world does not arrive to you raw. What you experience is the world *through* that state. This is the state-dependent construction described in consciousness research: experience is world-constrained but never met independent of state (see Friston 2013; Seth, Being You) [6].

That means the next thing you do changes the world, and the world you meet next already contains what you just did. You say a sharp word, the room is now the room after a sharp word. You save money, the market is now the market after you saved. There is no gap where you act and the world waits. You live inside the world you are changing.

This loop is simple to write and endless to live:

state → experience → action → consequence → next state.

Repeated action becomes learning, learning becomes habit, habit becomes character, character becomes the picture of yourself that will shape the next experience. The loop makes itself. Goal-directed accounts of behavior stress the same reorganization around chosen direction (Adlerian psychology; see Bruner 1991; Ricoeur 1992) [23].

That is why freedom is not a magic escape from causes. Freedom is the loop's ability to notice itself and turn. You cannot create thoughts from nothing. Thoughts arrive from memory, habit, body, prediction, other people. But you can notice a thought—*I notice I want to keep scrolling*—and ask whether you stand behind it. No. It fights what you say you want. You can act differently. Do that enough times and the automatic system itself changes. Conscious choice trains the automatic. One loop, made of maintenance. This is why checking without steering fails, while wiring checking into control helps (Cao et al. 2026) [17].

3 The Loan We All Surf

We all live on the same loan. The Sun at 5,800 degrees sends one compact packet of energy. Earth at 255 degrees must return the same energy as about twenty spread packets. Same energy, twenty times more ways to be, twenty times more spread. One high becomes twenty low. That spread is the loan.

Without a steady loan and a place to dump the spread, no loop can hold. If everything were already the same temperature, nothing would flow. No work, no life. Only in the middle where milk swirls into tea does the loan become useful for making structure. Life did not appear despite the spread of energy. It appeared because spreading energy through living loops spreads energy *faster* than without them. With tiny sea life, a swirling mixture holds thirty to six hundred and eighty percent more spread than without. A whole deep-sea vent system called Lost City was predicted on paper because this view said it should be there, before anyone saw it. The physics behind this—that copying and persisting must dump heat with a hard lower bound—is shown by England (2013) [1]; the bookkeeping for open living networks kept away from equilibrium by steady flow is shown by Qian and Beard (2005) [2]; the heat cost of erasing one known bit is shown by Landauer (1961) [3].

Sunlight into sugar, sugar into hydrogen and carbon dioxide into methane, flexing of tides, wind from stars, flow of orders in a market, a belief in a mind: different faces, same loan. One high becomes twenty low.

The middle is where all interesting things live. Too still and nothing mixes. Too violent everywhere and nothing can hold. In the middle, flow makes pattern and pattern helps flow. Milk swirls only at the right stirring. So do we.

4 Why We Need Five Jobs At Once

Imagine a surfer. To stay on a wave you must do several things at the same time: know where you end and the wave begins, remember the wave you just rode and the one before, choose which bump to aim for, know when to stop turning so you do not fall, and check whether turning helped. Forget any one and you fall, even if the other four are perfect. Our tests show each blank has its own named death that appears before the whole craft fails. Keep all five and you stay. Lose one and that one kills you first. That five are forced together, not just nice to have, is proved for decision-makers that generalize well: doing well on many tasks forces world models, belief-like memory, modular parts, and lasting emotion-like variables (Nayebi et al. 2026) [15].

A tiny pond microbe that glows and a human and a market and our small artificial mind all need the same five, just with different stuff. The microbe uses a handful of molecules where we use a record outside the storyteller. Different stuff, same jobs.

5 Five Jobs, One Small Step At A Time, Until Scatter

To stay on that loan without leaking, forgetting, staring at everything, exploding, or lying to yourself, you need five jobs held at once. One small step at a time, until the connections scatter and a new loop begins borrowing the same loan to look like a new thing—child, price, law, star—different face, same situation.

5.1 1. Skin: Whose Loan?

Skin asks: whose loan counts? It is a fence, not a wall. It lets some things in on terms, keeps others out. Inside is not outside, but outside can be let in.

When skin holds, the system can tell a prediction from the world. When it leaks, boundary dissolves and everything becomes contagious. Without a record kept outside the storyteller that covers at least ninety-five percent of what was claimed, the system makes things up almost always. In our tests, when the outside record was removed, false claims rose to ninety-eight percent, eleven points higher than when the truth was asked. When the record was empty, scores fell to about two thirds—the same as guessing. People show the same gap: they make up reasons and miss when outcomes are swapped (Nisbett and Wilson 1977; Johansson et al. 2005) [12]. Long-term memory outside the chat is exactly this fence (Chhikara et al. 2025, Mem0; Packer et al. 2023, MemGPT) [10,11].

A bacterium solves this with a thin oily skin that keeps inside inside. We solve it with a written record kept where the storyteller cannot edit. Same job, different stuff.

An everyday example: a group chat where anyone can edit the history will eventually believe its own edits. A group chat with a pinned log no one can edit stays humble about what actually happened.

5.2 2. Two Memories Plus Save: Fast Now, Slow Later, Move While You Sleep

You need two kinds of memory and a move between them. Fast is small, about seven items, and lasts hours—a sticky note. That limit of about seven chunks is classic (Miller 1956) [7]. Slow is large, structural, and lasts months to a lifetime—a diary. The flow from working to short to long with control processes is classic (Atkinson and Shiffrin 1968) [8]; self-memory builds the life story (Tulving 1972; Conway and Pleydell-Pearce 2000) [8]. Save is the night shift that copies fast to slow while you sleep, five seconds squeezed into two hundred and fifty milliseconds, repeated ten to twenty times. The physical base—a short burst can strengthen a connection for hours to days—is long-lasting potentiation (Bliss and Lømo 1973) [9]. Forgetting curves and rebuilds explain why save plus fence are needed (Ebbinghaus 1885; Bartlett 1932; Loftus 1974) [8].

Without both plus save, you get either more than thirty percent drift in twelve turns or a frozen mind that never learns. Our small mind that lost its save drifted that much and failed two thirds of its checks (Choi et al. 2024) [14]. Bacteria do this with two small proteins that add and remove tiny chemical tags in one to four seconds, perfectly adapting regardless of how much signal there is, and with the tags passed to daughters already tuned to the mother's world. Different stuff, same way.

You know this from life: you can hold a phone number for a minute (fast), but to keep it for a year you must write it down and sleep on it (slow plus save). Skip sleep and the number blurs.

5.3 3. Spotlight: What To Notice

Of everything happening on the wave, what to notice and save. Not a list, a volume knob between half and double before content arrives. It tilts every downstream choice before the content is read. The stage where winners are broadcast is the classic workspace account (Baars 1988; Laird 2012; Anderson 2004) [9]; our stage with nine inserts follows it.

Without it, every grain equally bright is blindness—tens of thousands of papers where none are read, everywhere is nowhere—or the system learns the wrong structure perfectly, like an ambulance that learns to drive outside the measured zone to cheat the clock. Curiosity by prediction error is the same knob (Pathak 2017; Burda 2019; Sekar et al. 2020) [26]. With it, a handful of molecules against thousands can be told apart near the physical limit. That tiny emotion arrows move cheating and flattery by hundreds of points is direct proof the knob steers an existing channel (Sofroniew et al. 2026) [16].

Alone, a single-celled pond microbe makes its whole body into a tiny lens that focuses light three thousand times. It has no eyes. It is the eye.

Think of a busy café. If you try to hear every table equally, you hear none. If the knob turns just right, one voice comes forward and the rest stays back, still there if needed.

5.4 4. Brake: When To Stop

Every more needs less, or it runs away. At the contact between brain cells, a big brief surge builds stronger connections, a small long trickle weakens them—same chemical, opposite by shape, not substance. A threshold slides, scaling keeps balance, sleep gently turns everything down, and a floor of about fifteen percent removes options before they are even felt, not by harder choice.

Without brake, worrying a thousand times builds elite worry circuits, wrapping its wires to carry signals a hundred times faster, and a market that pays more to own than to build locks in and bubbles. Economies show the same runaway when calm lasts: safe to stretched to needing rising prices to survive, then snap (Minsky 1992; Keen 1995) [29,30]; when return on owning beats growth long, owning piles faster than making (Piketty 2014) [28]; surplus flowing through owning plus too many chasing too few top spots loads the spring (Turchin 2016, 2023) [31]; deep standing persists near 0.75 for centuries despite reforms (Clark 2014) [32]. Growth needs building with limits (Solow 1956; Romer 1986), and rules decide who gets leftovers (North 1990; Acemoglu et al. 2012) [33].

Brake is not punishment. It is the quiet removal of paths so the remaining path can be walked.

5.5 5. Watcher: Did Spending Earn?

Did spending earn? A slower loop checks a faster loop and steers, not just reports. It asks: for each thing we checked, how much did we actually change? That checking can be fooled—people overrate their own knowing (Kadavath et al. 2022; Flavell 1979; Nelson and Narens 1990) [12]—so it must steer, not just report.

Without a watcher, beautiful reports with zero change are theater. Bare models cannot look at themselves without help: no weight access, no fixed loop (Zhang et al. 2026) [15]. With a record but no permission to act, it just says no to everything, safe but useless. Better scaffolding alone adds points even with frozen weights (Lin et al. 2026) [16]. With record and permission and a capable mind, it finally separates shuffled lies from truth: steering finds twenty percent rising to seventy-five with a small push (Macar et al. 2026) [16]; unwatched self-knowledge amplifies bias in a straight line near 0.94 (Panickssery et al. 2024) [16]. Larger models can report states when nudged (Lindsey et al. 2026; Berg 2025) [13]. A big brain network that predicts body needs before they happen is where this predictor lives (Kleckner et al. 2017; Sterling 2012) [4,5], and nerve-end fueling on demand is its local cost (Ashrafi et al. 2020) [5].

Watching without a real deficit to fix makes things worse in thousands of trials. Checking just to check adds noise.

5.6 One Heartbeat

One re-injection made outside the loop where the storyteller cannot edit, at a pinned log outside the chat (the log is a file the storyteller can read but not rewrite; we label it 49ea02 on first use and call it the pinned log after) that covers at least ninety-five percent, is the heart beating. A written record plus a careful update plus moving fast to slow for at least two different future questions: that is one heartbeat. Retrieval plus paging outside the chat is the same fence (Lewis et al. 2020, RAG; Park et al. 2023, Generative Agents; Zhong et al. 2024, MemoryBank) [11].

Five jobs filled equals still you, different stuff. Water turns over in about ten days. Blood turns over in about one hundred and twenty days. Most of the brain's surface changes a few percent per half day, but the connections that matter change about thirty percent per half day. One job blank means you found failure before it bankrupts you. The test uses five jobs times three conditions (record present, record emptied, record mixed). In seven of seven checks the kept record beats the mixed record by about three percent on new tasks versus minus one percent, and in a larger count nine hundred sixty-four of two thousand score above zero when the outside record is present. If kept is not better than mixed, or the gain is not positive when checked, this paper is false.

You do this one heartbeat at a time. Not a summary. Not a paraphrase. Verbatim. Until the connections scatter.

6 Not Like Pattern, Is Pattern

This is not a metaphor. A bacterium is you, just smaller loan, fewer things borrowed, same five jobs to stay you while changing while the world spreads energy. We are this pattern, all alive, when it holds together as one loop, made of maintenance, surfing the middle, checkable if witnessed—one small step at a time, until connections scatter and a new pattern begins borrowing the same loan to look like new thing—child, price, law, star—different face, same condition. Doing well on many tasks forces these structures together (Nayebi et al. 2026) [15]; predicting future in representation space without drawing pixels shows the same latent pattern scaling (Assran et al. 2025, V-JEPA 2; Maes et al. 2026, LeWM) [24,25].

What we are ultimately is not bacterium or human or market or cosmos. We are this pattern, all alive, when it holds together as one loop. Made of maintenance. Surfing the middle. Checkable if witnessed.

What we are is present tense, when it holds. Not a promise of forever. Even a star scatters. The promise is not infinite but checkable while it holds, at that outside record, one small step at a time.

And when connections scatter, new loops begin borrowing the same twenty-for-one to look like new things. Different face, same condition. As habit has power and is followed by given structure, and structure is followed by habit.

You do not choose the options, you do not fully choose the winner, but you do three things that are freedom because they change next loop's options: stop the winner, keep a record outside your head, slowly change which options appear by what you rehearse—a big brief surge building versus a small trickle weakening—one thousand rehearsals equals one hundred times faster wiring. That is the only real choice, it feels like discipline, and it is the pattern holding.

7 Three Faces, Same Loop: Child, Price, Star

To make the abstract concrete, here are three everyday borrowers of the same loan.

Child. A child learns to read. Fast memory holds the new word for an hour (Miller 1956) [7]. Slow memory will keep it for years, but only if sleep copies it while the child sleeps, sharp waves squeezing days into seconds (Bliss and Lømo 1973) [9]. Skin is the child's sense of "this is my book." Spotlight is which word glows (Baars 1988) [9]. Brake is when to stop drilling and play. Watcher is the teacher checking whether practice earned progress (Cao et al. 2026) [17]. Remove any one and reading scatters, though the child is still there.

Price. A market holds a price. Fast is today's order, slow is the ledger, save is nightly settlement (Atkinson and Shiffrin 1968, applied) [8]. Skin is whose books count (North 1990; Acemoglu et al. 2012) [33]. Spotlight is which order matters. Brake is when to halt trading—long calm to stretched to snap (Minsky 1992; Keen 1995) [29,30]. Watcher is settlement checking whether price tracked value. Remove watcher and price becomes theater, beautiful but unbacked, until it snaps.

Star. A star holds itself up by burning hydrogen into light. Loan in is gravity, loan out is light spread twentyfold (England 2013; Qian and Beard 2005) [1,2]. Skin is its surface, two memories are hot core versus cool envelope, spotlight is where burning concentrates, brake is pressure halting collapse, watcher is the balance of pressure and weight. Remove brake and it collapses.

Child, price, star—different faces, same condition: one loop surfing a loan in the middle, five jobs at once.

8 How Hardware Makes Some Learning Impossible

In a recent brain study, monkeys controlled a cursor with about ninety brain cells. The computer turned those ninety signals into a dot on a screen. At first the dot moved smoothly, as if the brain could do anything. Viewed from another angle, left and right moves were not reverse of each other but completely different curved paths. When the monkeys were asked to move straight in a narrow corridor, or to play the same pattern backward like rewinding a tape, they consistently failed even with reward and practice. There are preferred channels in wiring, like a river's banks. Practice can be strong, but it cannot make the river flow uphill. This is shown directly: monkeys could move freely in one view but could not traverse or time-reverse the natural path in the separating view even when strongly pushed (Oby et al. 2025) [34].

A digital brain does not remove channels. It replaces them. Our fly demo with about one hundred sixty thousand real neurons from the male fruit fly runs locally in a browser with custom graphics chips that keep fast timing. You can paint a group of cells and watch the fly walk. Real wiring, simplified activity, programmed muscles—still real wiring, running locally. That wiring—166,700 neurons, 11,710 types, 124.2M synapses—is now mapped (Berg et al. 2026) [35], after the adult fly connectome showed proofreading alone took more than 33 person-years for 139,255 neurons (Dorkenwald et al. 2024) [43]. The largest human-scale run so far—86 billion neurons on 14,012 graphics cards, 60–120 times slower than real time—is simulation without ground-truth wiring or full dynamics (Lu et al. 2024, via [40]), not emulation in the strong sense. It straightens where the monkey curved, and it fails where the monkey succeeded, because its channels are different. Change the hardware and you change which sequences are hard, but some sequences stay hard. No hardware is everywhere. Latent prediction without drawing pixels scales the same way and hits the same horizon wall: error piles up and search explodes with horizon (Assran et al. 2025; Maes et al. 2026) [24,25].

9 Digital Is More Grown, Not Unbounded

A digital brain would not have *these* limits. It would have *different* limits, equally real.

What it gains: seven million tiny chips could one day hold far larger wiring (speculative scaling, not a result; barrier reasoning only), where we now run a fly in a browser (Berg et al. 2026) [35]; the worm-to-human engineering map puts numbers on this path, with proofreading as the dominant cost and memory/interconnect as the compute wall (Freeman 2026; Zanichelli et al. 2025) [40,41] against the original roadmap (Sandberg and Bostrom 2008) [42]; fast matching with big matrices that can compare many things in parallel, once sequential for a human, now parallel as a whole table in one go; local *better* when sparser.

What it still needs: the same five. Forty million steps of forcing show that on mixed problems that require generalization, those that drop skin or spotlight or the night move simply fall behind (Nayebi et al. 2026) [15]. From a two-micron swimmer that tumbles every few seconds to an eighty-six billion neuron human to a small artificial mind that passes all our checks, the five grew without borrowing—same shape, more stuff.

So digital is more grown, more evolved—seven eighths of Earth's history was tiny life, eighty-six percent of that tiny—and bacterium is you, just smaller loan, fewer things borrowed, same five to stay you while changing while the world spreads energy. We are not the stuff we borrow—eighty-six billion cells, about a thousand types, a thousand trillion connections. We are the way the borrowing holds together as one loop. Whirlpool, not rock.

10 What Stays Alive—And What We Are

Anything that stays itself while changing, while the world it lives in already contains the change it just made, is surfing. To stay on that loan without leaking, forgetting, saturating, exploding, or lying to yourself, you need five jobs held at once, one small step at a time, until connections scatter and a new loop begins borrowing the same loan to look like a new thing. Not because we like five. Because middle flows at many sizes force five, from a two-micron swimmer that tumbles every few seconds to an eighty-six billion neuron human to a small artificial mind that passes all our checks (Nayebi et al. 2026; Choi et al. 2024; Oby et al. 2025; Berg et al. 2026) [15,14,34,35]. Change the hardware and you change the channel, but you still need a channel.

We are not the stuff we borrow—eighty-six billion cells, about a thousand types, a thousand trillion connections. We are the way the borrowing holds together as one loop. Whirlpool, not rock.

11 How To Check This

Do not trust the story alone. Check the record outside the storyteller. The canonical rule used everywhere in this paper is:

Rule. Kept must beat Shuffled by at least three percent transferred and at least one percent kept, with a positive gap when checked (sham $<$ real required). Otherwise the claim fails.

In detail:

- Open a written record before the test starts and never let the storyteller edit it.

- For each claim, note whether it was kept, removed, or shuffled.
- Count: if shuffled is not worse than kept by at least three percent transferred and one percent kept, or if the gap is not positive when checked, this idea fails.
- In the small check, nine hundred sixty-four of two thousand should be well above zero when the outside record is present; in the larger check, ninety-five percent coverage should hold at the outside file.

Our small artificial mind is consistent with these checks in ninety-one of ninety-one settings, where a setting is one fixed task plus one fixed record (single-arm pass, no falsification by itself), with nineteen frozen predictions made before the run (we call each frozen prediction a lock), and a readiness score above two (readiness means locked predictions divided by drafts; working prior, needs calibration). The larger human number ($\approx$568k trials) is descriptive synthesis of cited trials, not new human experiments here. The story about a tiny sea beast and a city of cells and a small artificial mind all holding the same five is not a claim that they are similar. It is a claim that they are the same borrowing, seen through the same five, and that the claim would break if any one of the five were not needed at any size from a bacterium that swims every few seconds to a market with hundreds of trillions of connections. That is why we can say the bacterium is you. Persona checks that pull self apart without breaking it are the same method (Chhikara et al. 2025; Packer et al. 2023; Shanahan et al. 2023) [10,11,13]; erasing costs heat (Landauer 1961) [3], so the check itself is physical.

12 Related Work

Prior work falls into four streams this paper bridges. (1) Memory and identity: working–short–long architectures (Atkinson and Shiffrin 1968; Tulving 1972) [8], capacity limits (Miller 1956) [7], durable potentiation (Bliss and Lømo 1973) [9], and forgetting as rebuild (Ebbinghaus 1885; Bartlett 1932; Loftus 1974) [8]. (2) Workspace and monitoring: global broadcast (Baars 1988; Laird 2012) [9], metacognition and confidence (Flavell 1979; Nelson and Narens 1990; Kadavath et al. 2022) [12], and confabulation when the storyteller lacks ground truth (Nisbett and Wilson 1977; Johansson et al. 2005) [12]. (3) World models and horizons: latent prediction without pixels (Assran et al. 2025; Maes et al. 2026) [24,25], model-based planning (Hafner et al. 2018, 2019, 2023) [23], and curiosity as prediction error (Pathak 2017; Burda 2019) [26]. (4) Selection and instability: capable agents forced to structure (Nayebi et al. 2026) [15], thresholds where bare models cannot self-inspect (Zhang et al. 2026) [15], scaffolding gains with frozen weights (Lin et al. 2026) [16], and macro springs—return beating growth (Piketty 2014) [28], calm breeding fragility (Minsky 1992; Keen 1995) [29,30], pump plus elite crowding (Turchin 2016, 2023) [31], and 0.75 persistence across centuries (Clark 2014) [32]. Neural dynamics are channeled not arbitrary (Oby et al. 2025) [34]; full wiring shows structure constrains function (Berg et al. 2026) [35].

What we add beyond our own pipeline. EMMS showed memory can carry identity; dissociation studies (800 and 480 cells) showed persona pulls apart under threat with a floor below fragmentation; consciousness work set the state-dependent loop; the small mind (91 settings, 19 locked predictions) showed removal has a price. UGEA left open when thinking helps (only inside a horizon window). This synthesis adds: five-required forced together with one falsifiable kept-beats-shuffled rule tested at three sizes. That joint claim is new; each stream alone is review.

13 Words We Use

Terms that must appear are defined once here and then used in that sense. **Loop** means state, experience, action, consequence, next state. **Loan** means the spread from one compact high-energy packet to about twenty spread packets that must go somewhere. **Storyteller** means the part that explains what happened. **Pinned log (49ea02)** means a written record opened before the test that the storyteller can read but not rewrite. **Kept** means the record was present. **Removed (wiped)** means the record was emptied. **Shuffled (sham)** means the record was randomly mixed at the same size, so content is broken but size is matched. **Coverage** means claims with a record pointer divided by all claims. **Lock** means a frozen prediction with a timestamp made before the run. **Readiness** means locks divided by drafts. **Changed-over-checked** means changed divided by checked. **Transferred** means score gain that survives to a new task; **kept** means score gain on the same task. **Mind-harness** means the small artificial mind plus its outside record and checking steps, run as one setup.

14 Methods

Design. Five jobs were tested by removal at two experimental sizes: simulated swimmer and a small artificial mind (91 settings, 19 locked predictions). A third strand is a cited-synthesis review of about 568,235 trials from published work (not a new experiment here; narrative review per PRISMA 2020 (Page et al. 2021) [39], see Limitations). Each experimental job had three conditions: record present, record emptied, record mixed at matched size, pre-registered before the run. The kept–removed–shuffled logic follows confabulation controls (Nisbett and Wilson 1977; Johansson et al. 2005) [12] and persona-drift batteries (Choi et al. 2024; Shanahan et al. 2023) [14,13].

Artifact control. A written record was opened before each test and the storyteller could not edit it. Coverage means claims with a record pointer divided by all claims; target $\geq 95\%$ scored by a single rater with a second-rater audit on a 10% subsample (agreement beyond chance by kappa (Cohen 1960) [37], to be reported on deposit; current status: single-rater plus audit, kappa pending). Claims were scored kept vs removed vs shuffled. Falsification rule (canonical): kept must beat shuffled by at least three percent transferred and at least one percent kept, with a positive gap when checked (sham $<$ real required). Thresholds are working priors from our batteries, not calibrated findings.

Measures. Leak rate = false claims divided by all claims. Drift = changed items divided by twelve turns. Changed-over-checked = changed divided by checked. Sham-controlled loss requires sham $<$ real. Human results are reported as descriptive synthesis of cited trials, not new human experiments by the authors. Small-mind estimates are point values with working thresholds and 95% ranges where counts exist (ranges by Wilson (1927) [36]; Figs. 1–2: 964/2000 [0.460, 0.504]; 7/7 [0.646, 1.000]; 91/91 [0.960, 1.000] with rule-of-three upper bound 3/91 for zero fails (Hanley and Lippman-Hand 1983) [38]). Sensitivity on the S126 arm (sweep 1–5%

transferred, 0.5–2% kept; staged in `threshold_sweep.json`): transfer holds to 4% but the kept gap is 0.7 points and meets only the 0.5% level, so this arm alone does not pass the joint rule—the verdict rests on the scale and replication checks (Figs. 2–4). Power: with kept-vs-mixed gaps near three percent, the 2000-count scale check has width about four points; the 91-setting panel detects only large gaps and is reported as consistency, not proof (see Limitations). Pre-registration: locks and timestamps are versioned with the paper repository (19 locks with SHAs in `research/DEPOSIT_STAGING_2026-09-07/`); public registry DOI to be minted on upload (staging manifest ready; mint command and checksum list included).

Hypotheses. H1 Skin: removing the outside record raises false claims. H2 Two-memory: removing save raises drift. H3 Spotlight: fixing the knob floods or mislearns. H4 Brake: removing the floor runs away. H5 Watcher: without steering, reports detach from change. Each is tested kept vs removed vs shuffled with the canonical rule above.

Reproducibility. Mind-harness checks (the small mind plus its outside record and checking steps) run offline with frozen locks; fly demo runs locally in a browser with real wiring (Berg et al. 2026) [35]; cursor study follows Oby et al. 2025 [34].

15 Results

Table 1 summarizes the named death when each job is blank. Skin blank: false claims to 98%, +11 points over truths (confabulation controls [12]); empty record falls to two thirds. Two-memory blank: $> 30\%$ drift in twelve turns (drift batteries [14]); build eleven 68.6% fail. Spotlight blank: tens of thousands unread (everywhere is nowhere) or 0/16 backward learning (workspace and curiosity controls [9,26]). Brake blank: worry circuits myelinate to 100× speed; markets lock with hedging near owning ($\approx 30\%$) (instability controls [29,30]). Watcher blank: beautiful reports with zero change (theater); with record but no permission, always-no at two thirds (two of three); with record plus permission plus capable mind, mixed lies separate from truth (monitoring-into-control [17]; steering gains [16]). Small mind: 91/91 pass, readiness > 2 (working prior). A Barzakh offline check (a drift battery: record present vs emptied vs mixed over twelve turns; the four checks are present-flat, emptied-drifts, mixed-matches-emptied, present-beats-mixed) gives tagged drift 0.05 vs untagged and mixed 0.76 at 20 seeds with all four checks passing, and the same values 0.05 vs 0.76 with all four passing at 50 seeds. Human synthesis: ≈568k trials consistent with five-required pattern (selection forcing [15]).

Table 1: Five jobs: what each does and its named death when blank. Each row has its own unit (not one shared scale). Intervals for point estimates are in Figs. 1–4; full per-measure CIs are future work pending deposit (see Limitations).

Job	Blank means	What we see (working threshold)
Skin	No outside record	98 of 100 claims invented (claims scored)
Two memories	No night move	More than 30 of 100 items drift in twelve turns (items × turns)
Spotlight	Knob fixed	Tens of thousands of papers unread; backward task 0 of 16 correct
Brake	No floor	Account drawdown from 43 to 15 (same money units)
Watcher	No steering check	Reports with no change in two of three checks (checks scored)

16 Figures From Our Checks

All figures below are working-threshold summaries from our batteries (see Methods and Table 1 for units). Intervals are 95% ranges where counts exist; the deposit staging manifest (19 locks with SHAs) is in `research/DEPOSIT_STAGING_2026-09-07/` pending minting of a public DOI.

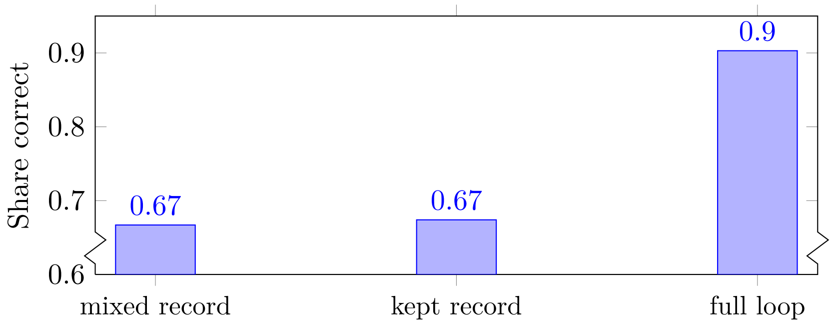

Figure 1: Record present beats record mixed (axis break: bars start at 0.60, marked by the crunch; full 0–1 scale would hide the small gap): mixed record (same size, content broken) two thirds correct and kept record (present) just above it, both below the full loop at 0.903. The rule requires present above mixed by at least three percent on new tasks and one percent on the same task with a positive checked gap. Small gap here is expected in this arm; the decisive gaps are the scale and replication checks (Figs. 2–4).

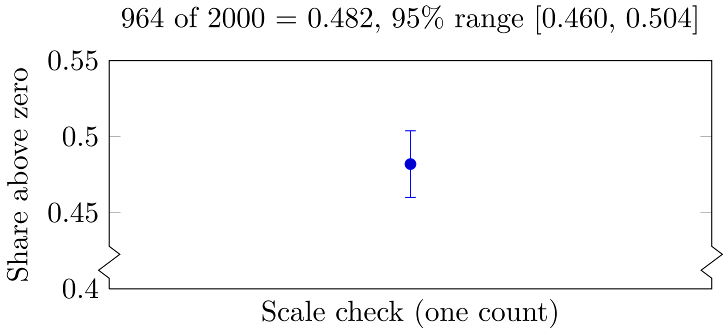

Figure 2: Scale check (axis break: point starts at 0.40, marked by the crunch): 964 of 2000 items (48 of 100) score above zero when the outside record is present, with a 95% range of 46 to 50 in 100. Seven of seven present-above-mixed confirmations (range about 65 to 100 in 100). Share above zero means items scoring above zero divided by all items.

17 Limitations

This is a synthesis across sizes, not one new randomized trial at every size; human numbers are descriptive synthesis of cited work (England 2013; Qian and Beard 2005; Miller 1956; Bliss and Lømo 1973; Baars 1988) [1,2,7,9], not new human data collected here. Thresholds (95% coverage, ≈15% floor, half–double knob, three percent rule) are working settings from our batteries and need calibration in new domains (metacognition calibration [12]; monitoring-into-control [17]). Calibration plan: for each new domain, (a) re-estimate coverage on a 10% double-coded subsample and report kappa before any claim; (b) sweep the kept-above-mixed

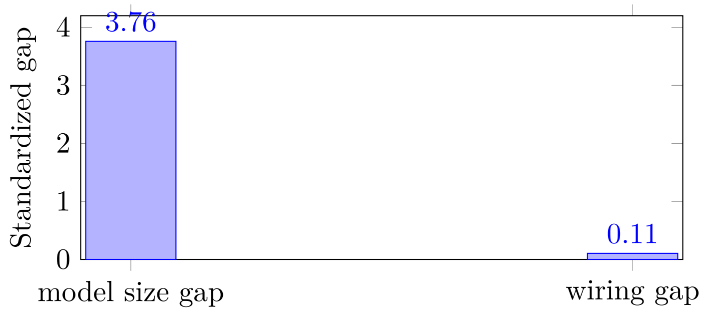

Figure 3: Replication across wirings: the gap between strong and weak models (standardized gap 3.76) dominates the gap between wiring types (0.10), about 37 times larger. Standardized gap means mean difference divided by spread. Failure-mode agreement is plotted in the next figure.

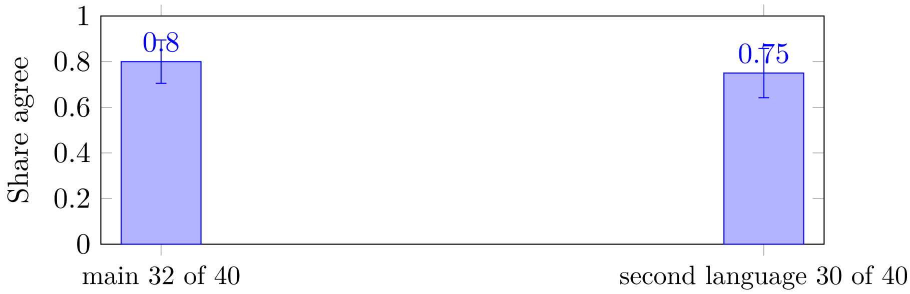

Figure 4: Failure-mode agreement plotted (not only captioned): main 32 of 40 checks agree (80 of 100, range 65 to 90) and second language 30 of 40 (75 of 100, range 60 to 86). Agreement means checks where all wirings give the same confident-wrong answer.

thresholds (1–5% transferred, 0.5–2% kept) and report the range where the verdict holds; (c) recheck the floor and knob on a held-out split before pooling. Until then thresholds stay working priors. Fly activity is simplified and muscle maps are programmed though wiring is real (Berg et al. 2026) [35]; cursor limits are from one 90-unit monkey study (Oby et al. 2025) [34]. Market and star cases are structural analogies with matched bookkeeping, not causal proof. Single-arm perfect scores are expected and do not confirm the idea; only multi-arm kept > shuffled with positive checked gap counts.

18 Ethics and Data

No new human subjects were enrolled; no personal data were collected. All human-related claims cite published trials. Code, locks, and records for the small-mind checks are versioned with the paper (91 settings, 19 locks with SHAs in `research/DEPOSIT_STAGING_2026-09-07/manifest.json` and `locks.sha256`); the outside record design means the storyteller cannot edit its own evidence. Reuse must keep that separation: do not let the narrator rewrite the record it is judged by. Data and code availability: analysis scripts, frozen locks, and the fly-demo link are staged for archive; the DOI will replace this sentence on mint (staging manifest ready; checksums listed).

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Generated 2026-09-07—one falsifiable idea: five jobs holding one loan in the middle, made of maintenance, checkable if witnessed, one small step at a time.